Initial Value Problems

Examples:

① Rigid pendulum

$$\theta''(t) = -\sin(\theta(t))$$
$$\theta(t_0) = \theta_0$$
$$\theta'(t_0) = \Omega_0$$

2nd-order ODE
Can be written as two first-order ODEs: $\Omega(t) = \theta'(t)$

$$\Rightarrow \quad \Omega'(t) = -\sin(\theta(t))$$

Because we can always do this, we will only focus on solving 1st-order ODEs.

Note also:

Given $u'(t) = f(u,t)$ non-autonomous

we can introduce $v = t$ with $v'(t) = 1$.

So we can consider only autonomous systems:

$$u'(t) = f(u)$$

② SIR Model (compartmental epidemiology)

β: contact rate

γ^{-1}: mean time of infectiousness

$S(t)$: Susceptible

$I(t)$: Infectious

$R(t)$: Removed

$$\boxed{S + I + R = 1}$$

$$\boxed{\begin{aligned} S'(t) &= -\beta S I \\ I'(t) &= \beta S I - \gamma I \end{aligned}}$$

$$R'(t) = \gamma I$$

$S(0) = S_0$

$I(0) = I_0$

$R(0) = R_0$

$$\frac{d}{dt}(S + I + R) = 0$$

Linear IVPs

Scalar, homogeneous:

$$u'(t) = \lambda u(t) \qquad \lambda \in \mathbb{C}$$
$$u(t_0) = \eta$$

$$u(t) = e^{\lambda(t-t_0)}\eta$$

Linear homog. system:

$$u'(t) = A u(t) \qquad A \in \mathbb{C}^{m \times m}$$
$$u(t_0) = \eta \qquad u : [t_0, T] \to \mathbb{C}^m$$

$$u(t) = e^{(t-t_0)A}\eta$$

$$e^M = \underline{I} + M + \frac{1}{2!}M^2 + \frac{1}{3!}M^3 + \cdots$$
$$= \sum_{j=0}^{\infty} \frac{1}{j!} M^j$$

Linear Inhom. System:

$$u'(t) = Au(t) + g(t)$$
$$u(t_0) = \eta$$

Duhamel's Principle:

$$u(t) = e^{(t-t_0)A}\eta + \int_{t_0}^{t} e^{(t-\tau)A} g(\tau)\,d\tau$$

Existence and Uniqueness

Linear IVPs:

A unique solution exists for all time.

Nonlinear IVPs: ?

Examples

$$u'(t) = (u(t))^2$$
$$u(0) = \eta > 0$$

$$u(t) = \frac{1}{\eta^{-1} - t}$$

Only defined for $t \in [0, 1/\eta)$.

$$u'(t) = \sqrt{u(t)}$$
$$u'(0) = 0$$

$$u(t) = \frac{t^2}{4}$$

or

$$u(t) = 0$$

Not unique.

Lipschitz Continuity

Given a function f on a domain D we say L is a Lipschitz constant for f on D if

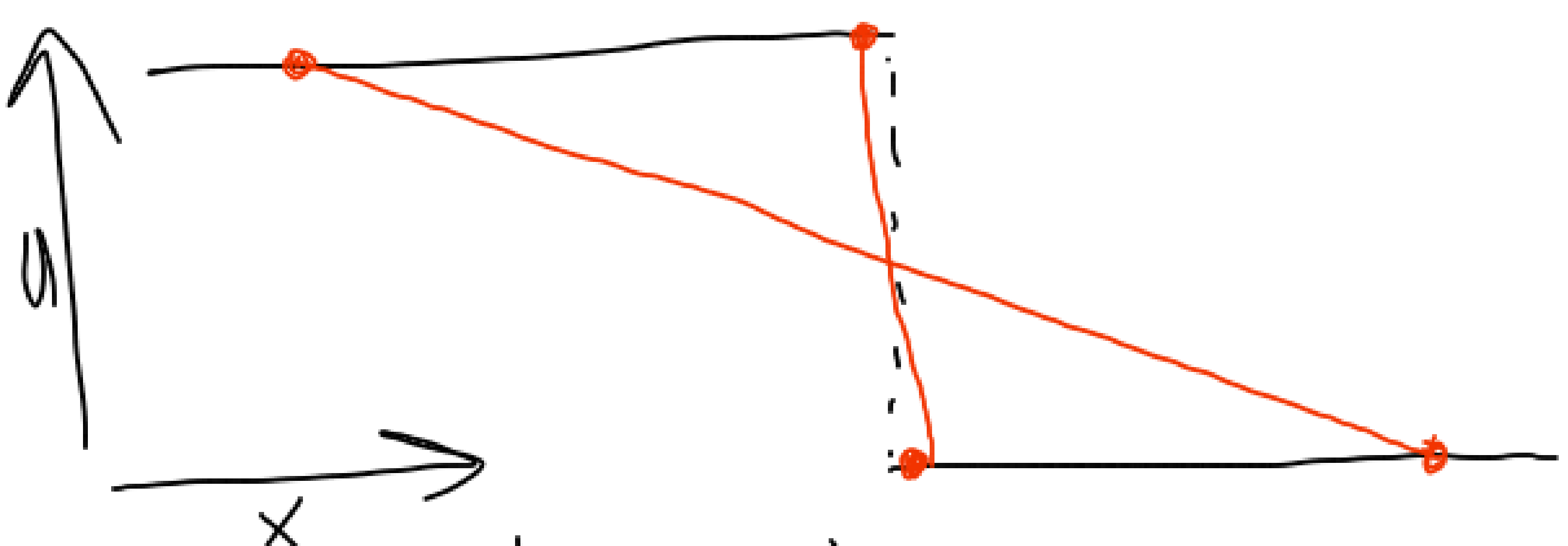
$$\|f(u) - f(v)\| < L\|u - v\|$$

for all $u, v \in D$ $(u \neq v)$

We say f is L. continuous if such an $L < \infty$ exists.

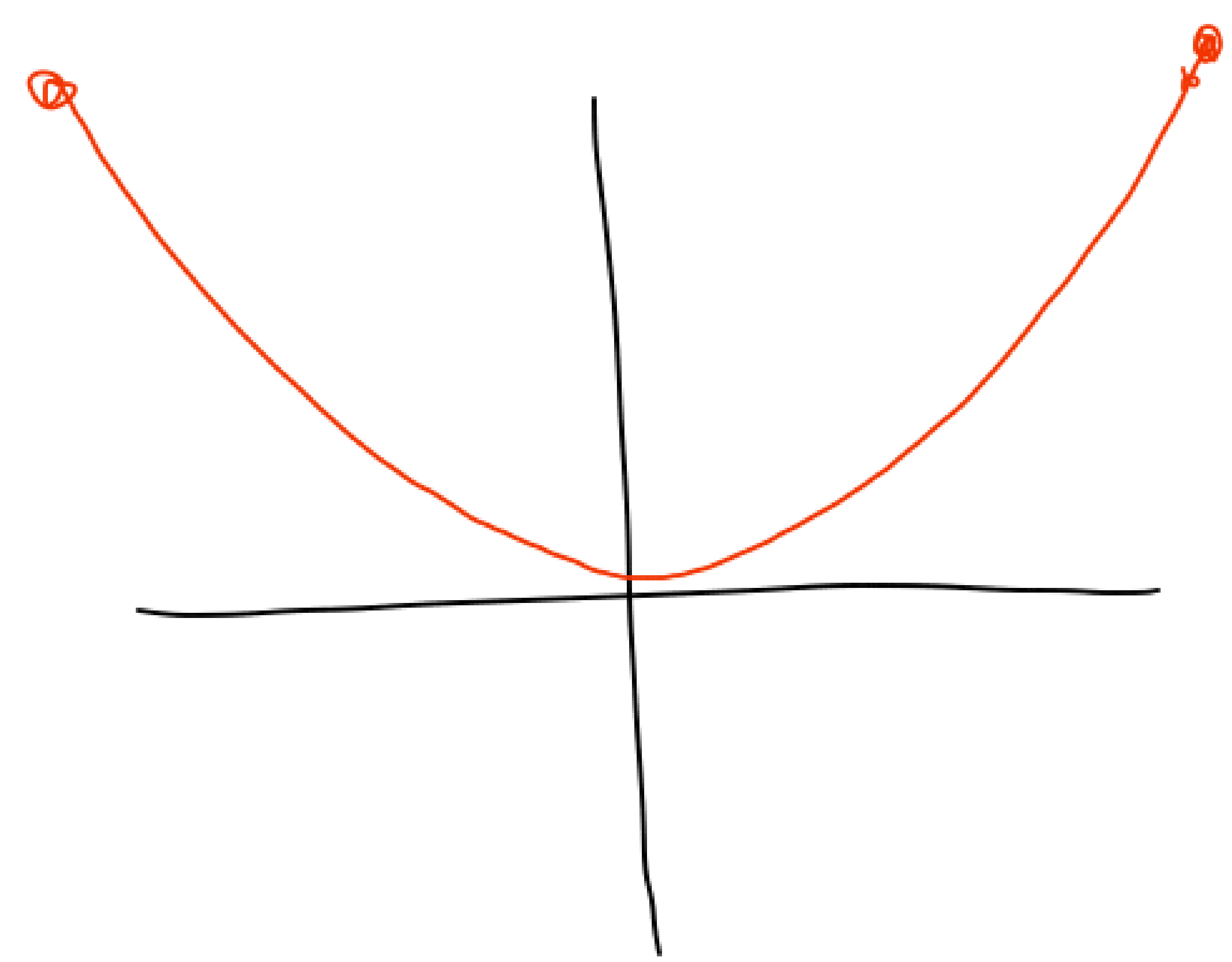
Heaviside function:

$$f(u) = \begin{cases} 0 & u<0 \\ 1 & u>0 \end{cases}$$



Not L. continuous.
(if $0 \in D$)

$f(u) = u^2$
$D: [-100, 100]$
$L = 100 + \varepsilon$



$f(u) = u^2$
$D: [0, \infty)$
Not L.C.

$f(u)=\sqrt{u}$

$D: [0, \alpha]$

Notice: $f'(u)$ unbounded as $u \to 0$.

$\Rightarrow$ Not L.C.

Given the IVP

$u'(t) = f(u)$ $\quad D = [\eta - a, \eta + a]$

$u(t_0) = \eta$

Suppose f is L.C. on D.

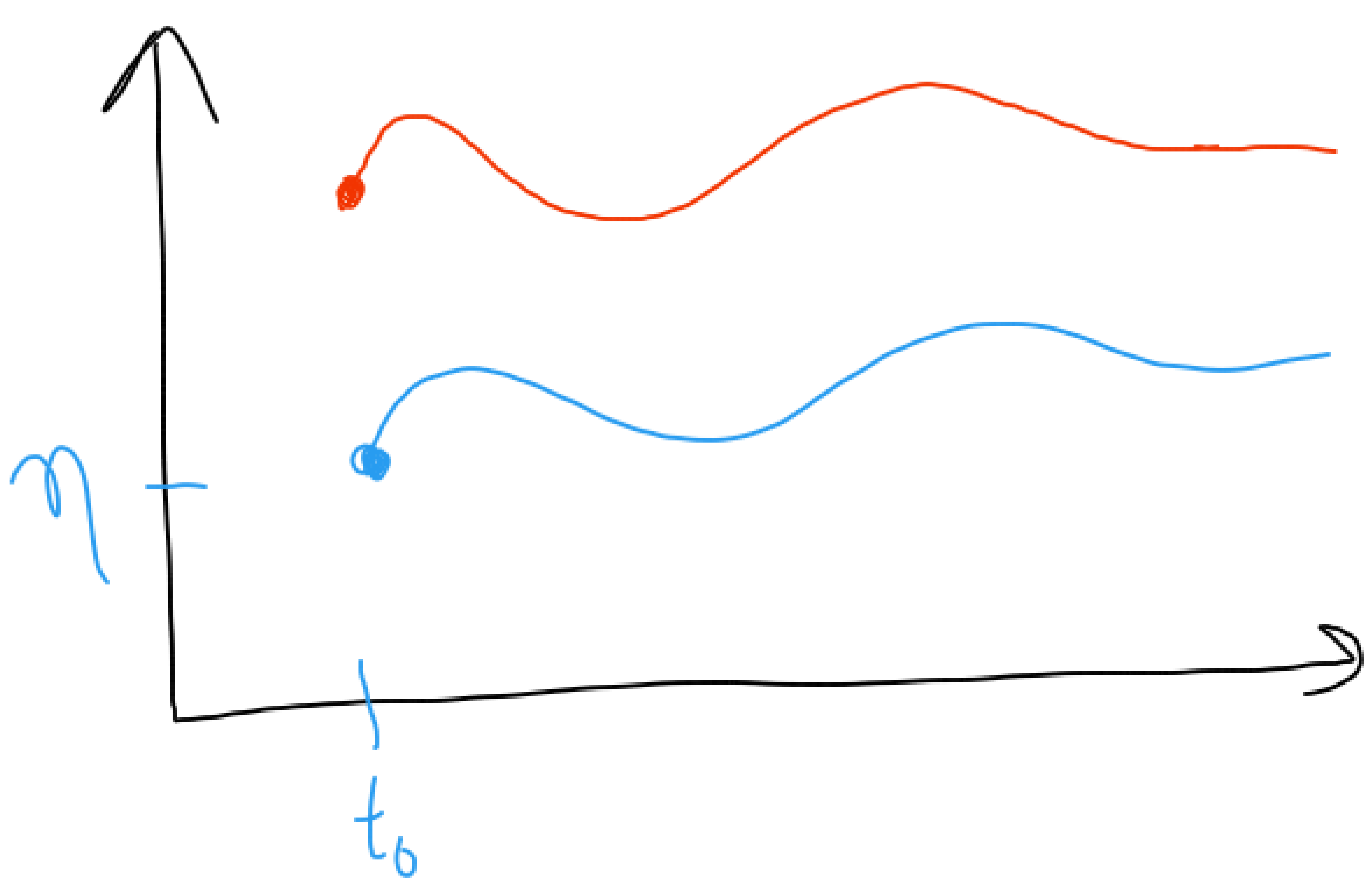
Then a unique solution exists

for $\quad t \le t_0 + \dfrac{a}{\sup_{u \in D} |f(u)|}$

Meaning of the Lipschitz Constant

① $u'(t) = g(t) \quad u(t_0) = \eta$

$u(t) = \eta + \int_{t_0}^{t} g(\tau)\,d\tau \quad L = 0$



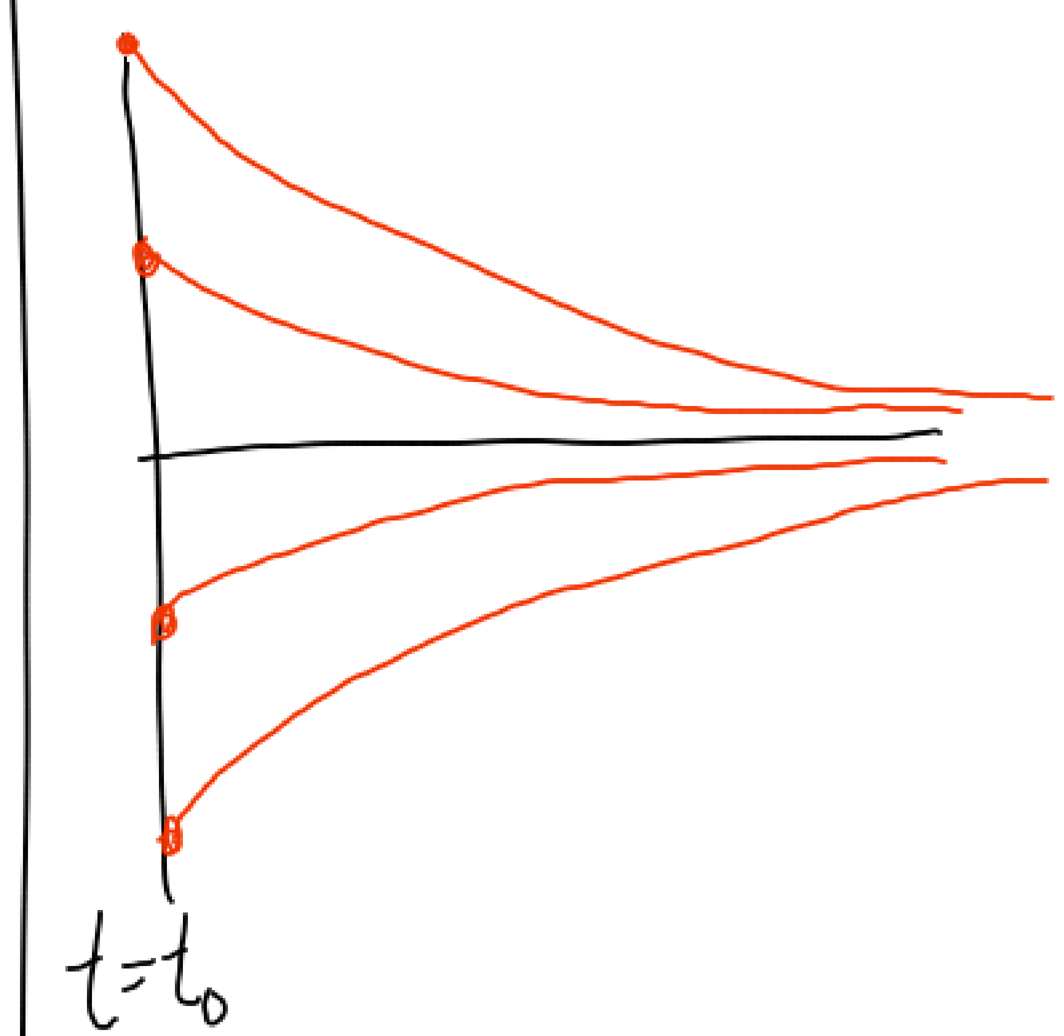
② $u'(t) = \lambda u(t)$

$u(t_0) = \eta$

$L = |\lambda|$

Large L: Solutions rapidly diverge or converge.

$\lambda < 0$:



$\lambda > 0$: